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Location: 20143 Milan, Italy



Alessandro Pedone

Curriculum Vitae

► PERSONAL DATA

Date of Birth	01/10/2002
Nationality	Italian
Place of Birth	Milan, Italy

► CURRENT STATUS

Politecnico di Milano	Pursuing Master's Degree in Mathematical Engineering, track in Computational Science and Computational Learning
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► EDUCATION

2024 - Current	Master's Degree in Mathematical Engineering, Politecnico di Milano
2021 - 2024	Bachelor's Degree in Mathematical Engineering, Politecnico di Milano Final grade: 110/110 cum laude Thesis: The Cauchy-Kowalevski theorem and some of its consequences Keywords: PDEs, characteristics method, analyticity/holomorphy, power series, method of majorants, Cauchy-Kowalevski, Holmgren and Cartan-Kähler theorems Supervisor: Prof. Maurizio Grasselli

► EXPERIENCE

01/12/2025 - 03/12/2025	Junior Math Days 2025 <i>Event organized every year by SISSA, with the goal of allowing Master's students in their final year to get in touch with the mathematical research environment (professors and PhD students) at SISSA.</i>
22/09/2025 - 26/09/2025	Workshop and Summer School on Applied Analysis 2025 <i>Organized by TU Chemnitz.</i> Topics: • High-Dimensional Approximations for Parametric and Random PDEs • Inverse Optimal Transport Introduction • Proximal Optimization and Deep Learning Methods (with applications to designing networks architectures) • Data-Driven Methods in Control (focus on Koopman-based techniques) • The Approximation Theory of (Shallow and Deep) Neural Networks

23/06/2025 - 27/06/2025

Theoretical Foundations of Machine Learning 2025

Admitted (as a Master's student) to a selective PhD program organized by MaLGa (Machine Learning Genoa Center) at the University of Genoa, with an acceptance rate of approximately 35%, primarily comprising PhD candidates.

Topics:

- Statistical learning theory framework
- Kernel methods and neural networks
- Empirical risk minimization and regularization
- Reproducing kernel Hilbert spaces (RKHS)
- Optimization techniques: convex analysis, gradient methods, stochastic optimization, splitting methods, backpropagation
- Theoretical analysis: concentration inequalities, empirical process theory, spectral calculus, operator theory, Rademacher complexities

► PROJECTS

04/2025 - Current

Geom-DeepONet (in progress)

Reference: [🔗](#)

Objective: develop a surrogate model for the electrostatic problem in an idealized MEMS with varying geometry.

Future applications: coupling the surrogate model to solve multiphysics (electro-mechanical) problems.

04/2025 - Current

Coupling MPE and Stokes' equations for cerebrospinal fluid flow and tissue motion (in progress)

Reference: [🔗](#)

Objective: study cerebrospinal fluid (CSF) flow, inside brain ventricles and subarachnoid space, with a focus on modeling the latter and exploiting already existing results.

Methodology:

1. *Implementation challenges*: segmenting the geometry (obtained from MRI) and creating the mesh of the subarachnoid space.
2. *Mathematical model*: coupling of MPE for brain tissue poromechanics Stokes' equations for cerebrospinal fluid flow.
3. *Numerical tools*: high order discontinuous Galerkin method on polytopal meshes for spatial discretization.

03/2025 - 05/2025

C++ Scientific Computing Projects

1. Some optimization methods for real-valued functions (Gradient Descent, Heavy Ball, Nesterov, ADAM...)
2. Parallel implementations of two kind of solvers for the Poisson equation in the unit square (Schwarz method)
3. Implementation of a matrix class, particularly suited for sparse matrices (CSR/CSC formats and modified formats)

12/2024

Mars Terrain segmentation

Objective: segment Mars terrain images into five classes (Background, Soil, Bedrock, Sand, and Big Rock) using a UNet architecture.

Dataset: 2,615 grayscale images (64x128 resolution), filtered to 2,505 images.

11/2024

Blood Cell Classification

Objective: perform an 8-class classification task on blood cells images using CNNs.

Dataset: 13,759 RGB images (96x96 resolution), filtered to 11,959 images after preprocessing.

04/2024 - 06/2024

Statistical Inference for Math Performance

Objective: build a linear regression model for math performance of italian high school students.

Dataset: OECD PISA 2022 survey.

► SKILLS

Programming Languages	Python, C/C++, MATLAB, R, LaTeX
Libraries	NumPy, SciPy, scikit-learn, TensorFlow, Keras, Pandas, Matplotlib, Seaborn, MIP, FEniCS, FEniCSx
Operating Systems	Linux (Ubuntu), Windows
Additional Software	Notion
Memory techniques	Method of loci, Mnemonic major system, Mnemonic link system

► LANGUAGES

Italian	Native speaker
English	C1 (IELTS 7.5)

► HONOURS AND AWARDS

25/01/2023	Premio Migliori Matricole aa 21-22 Classe L 8 Fondo Giovani Best Freshmen Award, academic year 2021–2022, Class L-8, Fondo Giovani — granted upon completion of the first year of enrollment, based on academic performance, including GPA and the number of ECTS credits earned.
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► MEMBERSHIPS

2025 - Current	SIAM (Society for Industrial and Applied Mathematics)
2021 – Current	AIM (Associazione Ingegneri Matematici)

► VOLUNTEERING

2020 - Current	Blood donor
2018	ABCDigital: Liceo Scientifico Elio Vittorini promoted a project to organize a free course for senior citizens, consisting of 10 weekly lectures (2 hours each), aimed at improving their digital skills, device usage, and online awareness.